Banach-Valued Sobolev Extension Sets: Verification, Current Status, and Further Partial Results

Research note prepared from the supplied partial solution

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Abstract

The supplied finite-dimensional argument is correct. This note gives a fully quantitative version: if a scalar Newton–Sobolev extension map has bound C_E, then an n-dimensional target V has an extension map with bound at most

$$2^{1-1/p}n^{1/p}C_E\,\|A\|\,\|A^{-1}\|, \qquad A : V \longrightarrow \ell^n_\infty.$$

The proof requires no Poincare inequality and works for $1 \le p < \infty$. A literature search did not locate a complete positive solution or a counterexample to Question 1.1 of Garcia-Bravo–Ikonen–Zhu. In fact, a May 2026 paper by the same authors explicitly states that even the Euclidean case $1 \le p < n$ remains open.

Two further infinite-dimensional positive results are established here in the Euclidean setting. If a scalar extension domain has a bounded linear scalar extension operator, then it is an extension domain for targets isomorphic to complemented subspaces of an L^p-space. In particular this covers L^p-targets, ℓ^p, and arbitrary Hilbert spaces. Finally, an explicit c_0-valued Lipschitz map on an interval shows why finite-rank approximation does not by itself pass the finite-dimensional argument to c_0: canonical truncations converge uniformly but not in Newton–Sobolev norm, and no finite-dimensional-range maps can converge to this example in that norm.

Bottom line. The partial theorem is valid after minor technical cleanup. No full answer or counterexample was found. The general problem remains open as of the search date, including the Euclidean range $1 < p < n$.

1 The question and its current status

Let $1 < p < \infty$, let Z be a p-PI metric measure space, and let $\Omega \subset Z$. Question 1.1 of Garcia-Bravo–Ikonen–Zhu asks whether every real-valued $W^{1,p}$-extension set is automatically a V-valued $W^{1,p}$-extension set for every Banach space V [1]. Their definition does not require an extension map to be linear.

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Question 1.1. *Let $1 < p < \infty$ and Z be a p-PI space. Are $W^{1,p}$-extension sets $\Omega \subset Z$ also $\mathbb{V}$-valued $W^{1,p}$-extension sets for all Banach spaces $\mathbb{V}$?*

More generally, does there exist a Banach space $\mathbb{V}_0$ such that every $\mathbb{V}_0$-valued $W^{1,p}$-extension set is a $\mathbb{V}$-valued $W^{1,p}$-extension set for all Banach spaces $\mathbb{V}$?

Many of the known real-valued extension results generalize to the Banach-valued setting with relative ease when the real-valued assumption is replaced by the corresponding Banach-valued one. The main difficulty in Question 1.1 arises in connecting the extensions properties for different Banach spaces: For example, if Ω is a real-valued $W^{1,p}$-extension set, an obvious approach to proving the existence of a, e.g. ℓ^∞-valued $W^{1,p}$-extension operator is to extend componentwise. However, without knowing that the extension operator commutes with projections, it is not obvious why the extension obtained in this way is even an $L^p(Z;\ell^\infty)$-function. Even if $L^p(Z;\ell^\infty)$-boundedness of the extension could be deduced (as one can do in many important cases, cf. Theorem 1.3 below), the Sobolev regularity of the extension is more difficult to deduce. We were able to make this idea work only in some special cases, cf. Section 1.4 below.

The 2023 paper proves several substantial positive cases. Among them:

- a c_0-valued extension set is an extension set for every Banach target;
- measure density together with a weak $(1, p)$-Poincare inequality up to some scale gives extension for every Banach target;
- Euclidean $W^{1,p}$-extension domains have the desired target-independent extension property when $p \geq n$;
- planar Jordan extension domains have it for all $1 < p < \infty$.

The most informative later source found in the search is the May 29, 2026 revision of *On infinity thick quasiconvexity and applications*, by the same three authors. It says explicitly that for domains in $\mathbb{R}^n$ the equivalence has been verified for $p \in [n, \infty)$, while $p \in [1, n)$ remains open [2]. Consequently, a complete answer to the original general p-PI question would in particular settle an open Euclidean subcase. No later full theorem or counterexample was located in searches by title, authors, citations, and the phrases "Banach-valued Sobolev extension set/domain" through June 19, 2026.

2 Conventions

We use the Newton–Sobolev convention of [1]. For a Banach-valued map u on a metric measure space X, write ρ_u for its minimal p-weak upper gradient and set

$$\|u\|_{W^{1,p}(X;V)} := \|u\|_{L^p(X;V)} + \|\rho_u\|_{L^p(X)}.$$

If $T : V \to W$ is bounded and linear, then

$$\|T \circ u\|_{L^p(X;W)} \leq \|T\| \, \|u\|_{L^p(X;V)}, \qquad \rho_{T\circ u} \leq \|T\| \rho_u \quad \text{a.e.}$$

The same upper-gradient conclusion holds for Lipschitz postcomposition.

An "extension map of bound C_E" means a map E satisfying $Eu|_\Omega = u$ and $\|Eu\| \leq C_E\|u\|$. If E is nonlinear, the phrase "operator norm" should be understood only as this Lipschitz-type bound at the origin.

3 Verification of the finite-dimensional result

Theorem 3.1 (Quantitative finite-dimensional extension)**.** *Let $1 \leq p < \infty$, let Z be a metric measure space, and let $\Omega \subset Z$ be measurable. Suppose there is a scalar extension map*

$$E_{\mathbb{R}} : W^{1,p}(\Omega) \longrightarrow W^{1,p}(Z)$$

with bound C_E. Let V be n-dimensional and let $A : V \to \ell_\infty^n$ be a linear isomorphism. Then there is a V-valued extension map E_V satisfying

$$\|E_V u\|_{W^{1,p}(Z;V)} \leq 2^{1-1/p} n^{1/p} C_E \|A\| \, \|A^{-1}\| \, \|u\|_{W^{1,p}(\Omega;V)}. \tag{1}$$

If $E_{\mathbb{R}}$ is linear, then E_V is linear.

Proof. We first take $V = \ell_\infty^n$. Write $u = (u_1, \ldots, u_n)$ and put

$$h_i = E_{\mathbb{R}} u_i, \qquad H = (h_1, \ldots, h_n).$$

Each coordinate projection is 1-Lipschitz, hence $u_i \in W^{1,p}(\Omega)$ and $\rho_{u_i} \leq \rho_u$ almost everywhere. In particular,

$$\|u_i\|_{W^{1,p}(\Omega)} \leq \|u\|_{W^{1,p}(\Omega;\ell_\infty^n)}. \tag{2}$$

Let $g_i = \rho_{h_i}$ and set $g = \max_{1 \leq i \leq n} g_i$. Outside the finite union of the exceptional p-negligible curve families for the g_i, every rectifiable curve γ satisfies

$$\begin{aligned} \|H(\gamma(a)) - H(\gamma(b))\|_{\ell_\infty^n} &= \max_i |h_i(\gamma(a)) - h_i(\gamma(b))| \\ &\leq \int_\gamma g \, ds. \end{aligned}$$

Thus g is a p-weak upper gradient of H. Strong measurability is automatic because the target is finite-dimensional, and the coordinate restriction identities imply $H|_\Omega = u$ almost everywhere.

Put

$$a_i = \|h_i\|_{L^p(Z)}, \qquad b_i = \|g_i\|_{L^p(Z)}.$$

Then

$$\begin{aligned} \|H\|_{W^{1,p}(Z;\ell_\infty^n)} &\leq \Big(\sum_i a_i^p\Big)^{1/p} + \Big(\sum_i b_i^p\Big)^{1/p} \\ &\leq 2^{1-1/p} \Big(\sum_i (a_i^p + b_i^p)\Big)^{1/p} \\ &\leq 2^{1-1/p} \Big(\sum_i (a_i + b_i)^p\Big)^{1/p} \\ &\leq 2^{1-1/p} C_E \Big(\sum_i \|u_i\|_{W^{1,p}(\Omega)}^p\Big)^{1/p} \\ &\leq 2^{1-1/p} n^{1/p} C_E \|u\|_{W^{1,p}(\Omega;\ell_\infty^n)}, \end{aligned}$$

where the last inequality uses (2).

For general V, define

$$E_V u = A^{-1} E_{\ell_\infty^n}(Au).$$

The bounded-linear postcomposition property gives

$$\|Au\|_{W^{1,p}(\Omega;\ell_\infty^n)} \le \|A\|\,\|u\|_{W^{1,p}(\Omega;V)}$$

and the analogous estimate for A^{-1} on Z. This proves (1). Linearity is inherited coordinatewise when the scalar extension map is linear. $\square$

Corollary 3.2 (A dimension-only bound)**.** *Every n-dimensional Banach space admits a linear isomorphism $A: V \to \ell_\infty^n$ with $\|A\|\,\|A^{-1}\| \le n$. Hence Theorem 3.1 always gives the bound*

$$2^{1-1/p} n^{1+1/p} C_E.$$

Proof. Choose an Auerbach basis $(v_i, v_i^*)_{i=1}^n$, so that $\|v_i\| = \|v_i^*\| = 1$ and $v_i^*(v_j) = \delta_{ij}$. The coordinate map $A(v) = (v_i^*(v))_{i=1}^n$ has norm at most one, while $\|v\| \le \sum_i |v_i^*(v)| \le n\|Av\|_\infty$. $\square$

What was right, and what needed tightening

The supplied packet had the correct central idea. The following are the only substantive cleanups.

1. No PI, doubling, completeness, or measure-density hypothesis is needed for the finite-dimensional statement.
2. The coordinate estimate is exact for the stated Newton–Sobolev norm; it is not merely true "up to an equivalent norm."
3. One should explicitly take the finite union of the exceptional curve families.
4. The extension is linear exactly when the original scalar extension is linear.
5. Formula (1) records the quantitative loss.

The same finite-coordinate mechanism also appears in Lemma 4.5 of [1], there in a density argument rather than as an extension theorem.

4 A further infinite-dimensional positive result in Euclidean space

The next result does not solve the open problem, but it goes beyond finite dimension. Its key point is that the Sobolev graph norm tensorizes exactly with an L^p target when the target exponent matches the Sobolev exponent.

For an open $D \subset \mathbb{R}^n$ and a Banach space V with the Radon–Nikodym property, use the equivalent classical graph norm

$$\|u\|_{\mathcal{W}^{1,p}(D;V)} := \left(\|u\|_{L^p(D;V)}^p + \sum_{j=1}^n \|\partial_j u\|_{L^p(D;V)}^p \right)^{1/p}. \tag{3}$$

On Euclidean open sets, the Newton–Sobolev and Reshetnyak spaces agree for all Banach targets, and the classical Bochner–Sobolev space agrees with them precisely for targets having the Radon–Nikodym property [3].

Lemma 4.1 (Sobolev–Fubini identification)**.** *Let (S,ν) be a σ-finite measure space and $D \subset \mathbb{R}^n$ be open. Under the identification $U(s)(x) = u(x)(s)$,*

$$\mathcal{W}^{1,p}(D; L^p(S,\nu)) \cong L^p\big(S,\nu; \mathcal{W}^{1,p}(D)\big)$$

isometrically for the norm (3).

Proof. For a Bochner–Sobolev map u, Fubini's theorem and the distributional definition of the weak partial derivatives show that for almost every s, the slice $U(s)$ lies in $\mathcal{W}^{1,p}(D)$ and $\partial_j U(s) = (\partial_j u)(\cdot)(s)$. Conversely, the same calculation against tensor-product test functions gives the weak derivatives of u from those of the slices. Finally,

$$\|u\|^p_{\mathcal{W}^{1,p}(D;L^p(S))} = \int_S \left(\|U(s)\|^p_{L^p(D)} + \sum_{j=1}^n \|\partial_j U(s)\|^p_{L^p(D)} \right) d\nu(s),$$

which is the asserted isometry. □

Theorem 4.2 (L^p-target tensorization)**.** *Let $1 < p < \infty$, let $\Omega \subset \mathbb{R}^n$ be open, and suppose there is a bounded linear scalar extension operator*

$$E : \mathcal{W}^{1,p}(\Omega) \longrightarrow \mathcal{W}^{1,p}(\mathbb{R}^n).$$

Then, for every σ-finite (S, ν), there is a bounded linear extension operator

$$E_{L^p(S)} : \mathcal{W}^{1,p}(\Omega; L^p(S,\nu)) \longrightarrow \mathcal{W}^{1,p}(\mathbb{R}^n; L^p(S,\nu))$$

with the same operator bound when both spaces use the graph norm (3)*.*

Proof. By Lemma 4.1, an $L^p(S)$-valued Sobolev map corresponds to a Bochner map

$$U \in L^p\big(S; \mathcal{W}^{1,p}(\Omega)\big).$$

Every bounded linear map between Banach spaces acts pointwise, with the same norm, on the corresponding Bochner L^p spaces. Define

$$\widetilde{E}U(s) := E(U(s)).$$

Then

$$\|\widetilde{E}U\|_{L^p(S;\mathcal{W}^{1,p}(\mathbb{R}^n))} \le \|E\|\, \|U\|_{L^p(S;\mathcal{W}^{1,p}(\Omega))}.$$

Transporting $\widetilde{E}U$ back through Lemma 4.1 gives the desired extension. The scalar restriction identity holds for almost every s, hence the resulting $L^p(S)$-valued map restricts to the original map. □

Corollary 4.3 (Complemented subspaces of L^p)**.** *Under the hypotheses of Theorem 4.2, let V be isomorphic to a complemented subspace of some $L^p(S,\nu)$. Then Ω is a V-valued extension domain. Quantitatively, if $J : V \to L^p(S,\nu)$ is an isomorphic embedding and $Q : L^p(S,\nu) \to J(V)$ is a bounded projection, the extension bound is at most*

$$\|J^{-1}\|\, \|Q\|\, \|E\|\, \|J\|.$$

Proof. Apply Theorem 4.2 to $J \circ u$, then postcompose the extension with $J^{-1}Q$. □

Corollary 4.4 (Concrete targets)**.** *Let $1 < p < \infty$ and let $\Omega \subset \mathbb{R}^n$ be a scalar $W^{1,p}$-extension domain. Then Ω is a Newton–Sobolev extension domain for each of the following infinite-dimensional targets:*

$$L^p(S,\nu), \qquad \ell^p, \qquad \textit{and every Hilbert space } H.$$

More generally, it works for every Banach space isomorphic to a complemented subspace of an L^p-space.

Proof. A Euclidean scalar extension domain satisfies the measure-density condition [4]. The linearization theorem in [1] therefore provides a bounded linear scalar extension operator, so Theorem 4.2 and Corollary 4.3 apply.

It remains only to recall why Hilbert spaces are covered. Let $H = \ell^2(I)$ and let $(r_i)_{i\in I}$ be independent Rademacher functions on $\{-1,1\}^I$ with product probability measure. Khintchine's inequality gives an isomorphic embedding

$$J : \ell^2(I) \longrightarrow L^p(\{-1,1\}^I), \qquad Ja = \sum_{i\in I} a_i r_i.$$

For $f \in L^p$, set $c_i = \int f r_i$. By duality and Khintchine's inequality for $p' = p/(p-1)$,

$$\|(c_i)\|_{\ell^2(I)} \le C_{p'} \|f\|_{L^p}.$$

Thus

$$Qf := \sum_{i\in I} c_i r_i$$

defines a bounded projection onto the Rademacher span. Hence H is isomorphic to a complemented subspace of an L^p-space. Such targets are reflexive, hence have the Radon–Nikodym property, so the classical and Newton–Sobolev formulations agree by [3]. □

Remark 4.5. Corollary 4.4 is not a solution for arbitrary Banach targets. The tensor extension of a general bounded scalar operator need not be bounded on $L^p(\cdot; V)$ for an arbitrary V; complemented L^p structure is the precise mechanism used above.

5 Why finite-rank approximation does not finish the problem

The source paper proves energy-density for targets with the metric approximation property but deliberately does not claim norm-density. The following elementary example shows a sharp obstruction.

Proposition 5.1 (c_0 truncations can fail completely)**.** *Let $I = (0,1)$ and define*

$$u(t) = \left(\frac{\sin(kt)}{k}\right)_{k=1}^{\infty} \in c_0.$$

For every $1 \le p < \infty$:

(i) *u is 1-Lipschitz, hence belongs to the Newton–Sobolev space $W^{1,p}(I; c_0)$;*

(ii) *if P_N is the first-N coordinate projection, then $P_N u \to u$ uniformly (and hence in L^p), but*

$$\rho_{u-P_N u} = 1 \quad \textit{a.e. on } I;$$

in particular $P_N u$ does not converge to u in $W^{1,p}(I; c_0)$;

(iii) *u is not in the classical Bochner–Sobolev space $W^{1,p}_{\mathrm{Bochner}}(I; c_0)$;*

(iv) *no sequence of Newton–Sobolev maps whose essential ranges lie in finite-dimensional subspaces of c_0 can converge to u in Newton–Sobolev norm.*

Proof. Every coordinate $t \mapsto \sin(kt)/k$ is 1-Lipschitz, so the supremum norm gives $\|u(t)-u(s)\|_{c_0} \le |t-s|$. Also $|\sin(kt)|/k \to 0$, so the image lies in c_0.

For $v_N = u - P_N u$,

$$\|v_N\|_{L^\infty(I;c_0)} \le \frac{1}{N+1}.$$

On the other hand, for every t,

$$\sup_{k>N} |\cos(kt)| = 1.$$

Indeed, this is immediate when $t/(2\pi)$ is rational by taking multiples of a denominator, and follows from density of the irrational rotation when $t/(2\pi)$ is irrational. The metric derivative therefore satisfies

$$\operatorname{md} v_N(t) = \lim_{h\to 0} \frac{\|v_N(t+h) - v_N(t)\|_{c_0}}{|h|} = 1.$$

The upper bound follows from 1-Lipschitz continuity, while the lower bound is obtained by fixing a coordinate with $|\cos(kt)|$ arbitrarily close to one. On an interval, the minimal weak upper gradient of a Lipschitz metric-valued map equals its metric derivative almost everywhere, proving (ii).

If a Bochner derivative $u'(t) \in c_0$ existed, every coordinate projection would give

$$u'(t) = (\cos(kt))_{k=1}^{\infty}.$$

This sequence is not in c_0 for any t, so u is nowhere Bochner differentiable. This proves (iii); it is the standard example recorded in [3].

Finally, on Euclidean open sets the Newton–Sobolev and Reshetnyak spaces coincide, while the classical Bochner–Sobolev space is a closed subspace of the Reshetnyak space [3]. Every finite-dimensional-range Newton–Sobolev map is Bochner–Sobolev, because finite-dimensional targets have the Radon–Nikodym property. Norm convergence of such maps to u would therefore force u into that closed classical subspace, contradicting (iii). □

Remark 5.2. The proposition does not produce a counterexample to the extension question: I itself is already a PI space and u needs no extension. It rules out a natural proof strategy. One cannot pass from the finite-dimensional theorem to c_0 merely by truncating the target and taking a Newton–Sobolev norm limit.

6 Strategies tested against the remaining problem

The following approaches were examined; each isolates a genuine obstruction.

Coordinatewise extension to c_0 or ℓ^∞. Finite coordinates give Theorem 3.1, but the bound grows with the number of coordinates. For infinitely many coordinates, scalar bounds do not ensure that the assembled map is Bochner measurable, L^p-bounded, or controlled by one common upper gradient. Proposition 5.1 shows that finite-coordinate limits need not converge in Sobolev norm.

Metric approximation property. Finite-rank operators approximate compact subsets of the target and yield the energy-density theorem in [1]. Energy convergence is weaker than Newton–Sobolev norm convergence; Proposition 5.1 shows that the gap is not a technicality.

Scalarization by the dual unit ball. For every $v^* \in B_{V^*}$, the scalar function $v^* \circ u$ extends. The problem is to choose these scalar extensions compatibly. Uniform scalar norm bounds do not produce a single pointwise L^p upper gradient valid for all v^*, nor do they guarantee that the values come from one element of V rather than merely from V^{**} or a product space.

Linear tensor extension. This succeeds for complemented subspaces of L^p by Section 4, and for Hilbert targets through the Rademacher complement. For arbitrary V, a bounded scalar operator on an L^p-type graph subspace need not have a bounded amplification by Id_V. Thus ordinary boundedness of the scalar extension operator is not enough for abstract tensorization.

A nonlinear-Poincare or expander counterexample. One might try to encode a family with good scalar Poincare inequalities but bad c_0-valued inequalities. Classical expanders provide that functional- analytic contrast, but an unbounded expander family is not uniformly doubling. Since every subset of a doubling metric space is metrically doubling, this mechanism cannot be inserted at uniform scale into a fixed PI ambient space without losing the hypotheses. No valid PI realization was found.

Removable sets and zero-capacity bottlenecks. If the removed set has zero p-capacity, the family of curves meeting it is p-negligible, a target-independent fact. Such examples therefore tend to extend all metric or Banach targets, rather than separate scalar and vector extension. Positive-capacity bottlenecks usually already obstruct scalar extension.

These failures are consistent with the formulation in [1]: in the measure-dense, path-almost-open setting, the open issue can be reduced to a closed-image/completeness question for the c_0-valued local Hajlasz–Sobolev space. None of the arguments above establishes or refutes that completeness.

7 Conclusion

The supplied partial result is mathematically sound and can be strengthened to the quantitative Theorem 3.1. The literature does not currently supply a full answer or counterexample; the same authors still identify the Euclidean range $1 < p < n$ as open in May 2026. The tensorization argument above adds a genuine infinite-dimensional class in Euclidean space—all targets complemented in an L^p-space, including L^p, ℓ^p, and Hilbert spaces—but arbitrary Banach targets remain beyond it. The c_0 example explains why the most direct finite-rank limiting argument cannot bridge the remaining gap.

References

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